MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, CLIPPER, ANALOG, KEY CODE FBE

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic clipper amplifier module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

	Name	Amplifier, clipper
	Family	Analog
	Key code	FBE
	Size	1A
	Weight	35 grams maximum
	Failure rate	0.7 failures/10^6 hours maximum
*	Environment	Class I as specified in MIL-STD-1389
*	Maximum power dissipation +15 Vdc supply -15 Vdc supply 5 Vdc supply	1.76 W (no signal) 825 mW 825 mW 105 mW
	Supply voltages	5 Vdc ±5 percent and ±15 Vdc ±10 percent
	Input voltage Normal Abnormal	 0.5 mVrms to 3 Vrms white noise ±15 V
	Output voltage Low level High level	 0.0 V minimum, 0.5 V maximum 2.5 V minimum, maximum $\leq V_{CC} \leq$ 5.5 V
	Output drive Low level (V_{OL} = 0.5 V) High level (V_{OH} = 2.4 V)	 10 mA maximum 1.0 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN. ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Propagation delay time	0.200 ±0.100 µs
Small signal bandwidth	500 kHz nominal
Output voltage symmetry	±1 percent maximum
Transition time (t_{TLH}) C_L = 15 pF	18 ns minimum, 70 ns maximum
(t_{THL}) C_L = 15 pF	5 ns minimum, 70 ns maximum
Output noise	-40 dbV/$\sqrt{Hz}$ (20 Hz to 50 kHz)
Crosstalk	±1 percent of unity correlation

* 1.3 Module function. This module contains a very high gain linear amplifier, with back-to-back diode feedback, followed by a voltage comparator that converts the output to a logic level.

1.3.1 Parallel outputs. Paralleling of outputs is not permissible.

* 1.3.2 Ground connections. To insure proper module operation a unipoint ground as shown on figure 9 is recommended.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.7 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of the design class I as specified in MIL-STD-1389.

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3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FBE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20. Contact pins in positions A21 through A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/233-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.00 ± 0.25 Vdc and ±15.0 ±1.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.25 Vdc ±1 percent and 16.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P07	5
Sine wave generator	Item G01	1
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y02	2
Oscilloscope probe	Item Y05	1
Oscilloscope	Item D06	1
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	1
DC ammeter	Item X01	3
Digital multifunction meter	Item M02	2

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* 4.3 Test equipment - continued.

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure
Wave analyzer	Item B03	1
Ohmmeter	Item R01	1
Noise generator	Item K03	2
Resistors		
600Ω ±5%, 1 W		2
499Ω ±1%, 1/8 W		1
51Ω ±5%, 1 W		2
499Ω ±1%, 1/4 W		1
1 kΩ ±1%, 1/4 W		2
100Ω ±0.1%, 1/4 W		1
100 kΩ ±0.1%, 1/4 W		2
2.43 kΩ ±1%, 1/8 W		1
Value as required for figure 9, note 4, 1/8 W		1
2 kΩ ±1%, 1/8 W		1
1 kΩ ±1%, 1/8 W		2
4.7 kΩ ±1%, 1/4 W		1
10 kΩ ±1%, 1/8 W		1
105 kΩ ±1%, 1/8 W		1
10 kΩ, 1/8 W potentiometer		1
26.1 kΩ ±1%, 1/8 W		1
100Ω ±1%, 1/8 W		1
49.9Ω ±1%, 1/8 W		2
49.9 kΩ ±1%, 1/8 W		2
15 kΩ ±1%, 1/8 W		6
50 kΩ ±1%, 1/8 W		4
Capacitors		
100 pF ±20%		1
10 µF ±20%, 50 V, nonpolar electrolytic		1
Value as required for figure 9, note 5		1
0.1 µF ±10%		1
5 µF ±1%, 20 V, electrolytic		1
2,000 pF ±20%, 50 V		4
Diode		
1N916, or equivalent		1
Transistors		
2N2907A, or equivalent		1
2N2222A, or equivalent		1
Integrated circuits		
LM311, or equivalent		1
SN7474, or equivalent		1
SN7408, or equivalent		1
µA741, or equivalent		1
HA2600, or equivalent		2
Switches		
SPST		1
SPDT		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through L, M2, M3, N, and R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition N, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, and while still at the maximum class temperature, electrical tests E through L, M2, M3, N, and R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module temperature to 25°C, electrical tests D through R shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/233-1 supersedes part number M28787/233. The two parts are interchangeable.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

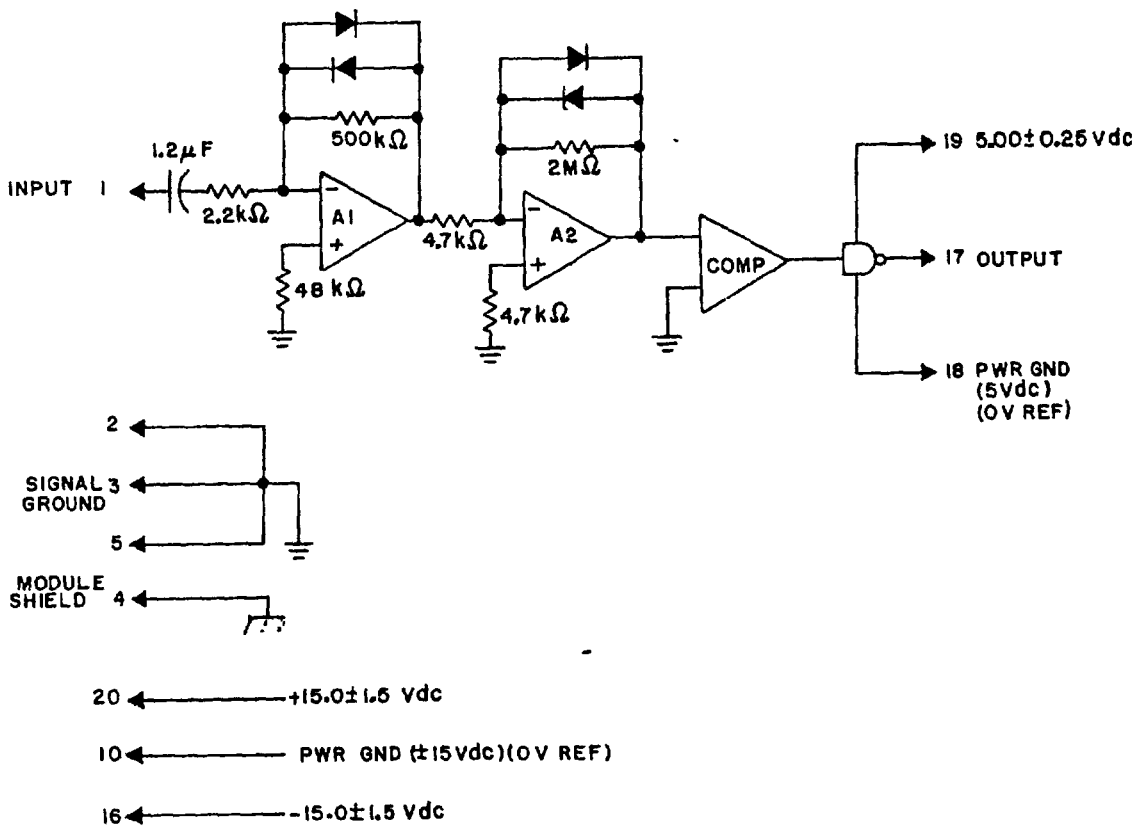
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NOTES:
1. All component values are for reference only.
2. Pins 6 through 9 and 11 through 15 are unused.

FIGURE 1. Module function diagram, clipper amplifier

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	V_S stress output signal	II	Q	900	1100	900	1100	900	1100	900	1100	Hz
B	Output short circuit current	III	Q	20	100	20	100	20	100	20	100	mA
C	Recovery time	IV	S	0	100	0	100	0	100	0	100	ms
D1-D11	Isolation	V	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
E1,E3	Output signal,voltage, low level	VI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
E2,E4	Output signal,voltage, high level	VI	X	2.5	5.0	2.5	5.0	2.5	5.0	2.5	5.0	V
E5	Output signal, 10 kHz	VI	X	96	104	96	104	96	104	96	104	µs
E6	Output signal,small signal bandwidth check	VI	S	1.8	2.2	1.8	2.2	1.8	2.2	1.8	2.2	µs
F1	Transition time (t_{TLH})	VII	X	18	70	18	70	18	70	18	70	ns
F2	Transition time (t_{THL})	VII	X	5	70	5	70	5	70	5	70	ns
F3	Transition time (t_{TLH})	VII	Q	30	100	30	100	30	100	30	100	ns
G1-G2	Propagation delay time (t_{PLH} and t_{PHL})	VII	X	100	300	100	300	100	300	100	300	ns
H1-H2	Power supply current, ±15 V	VIII	X	10	50	10	50	10	50	10	50	mA
H3	Power supply current, 5 V	VIII	X	2	20	2	20	2	20	2	20	mA
J	Input impedance check	IX	S	0.50	1.01	0.50	1.01	0.50	1.01	0.50	1.01	Vrms

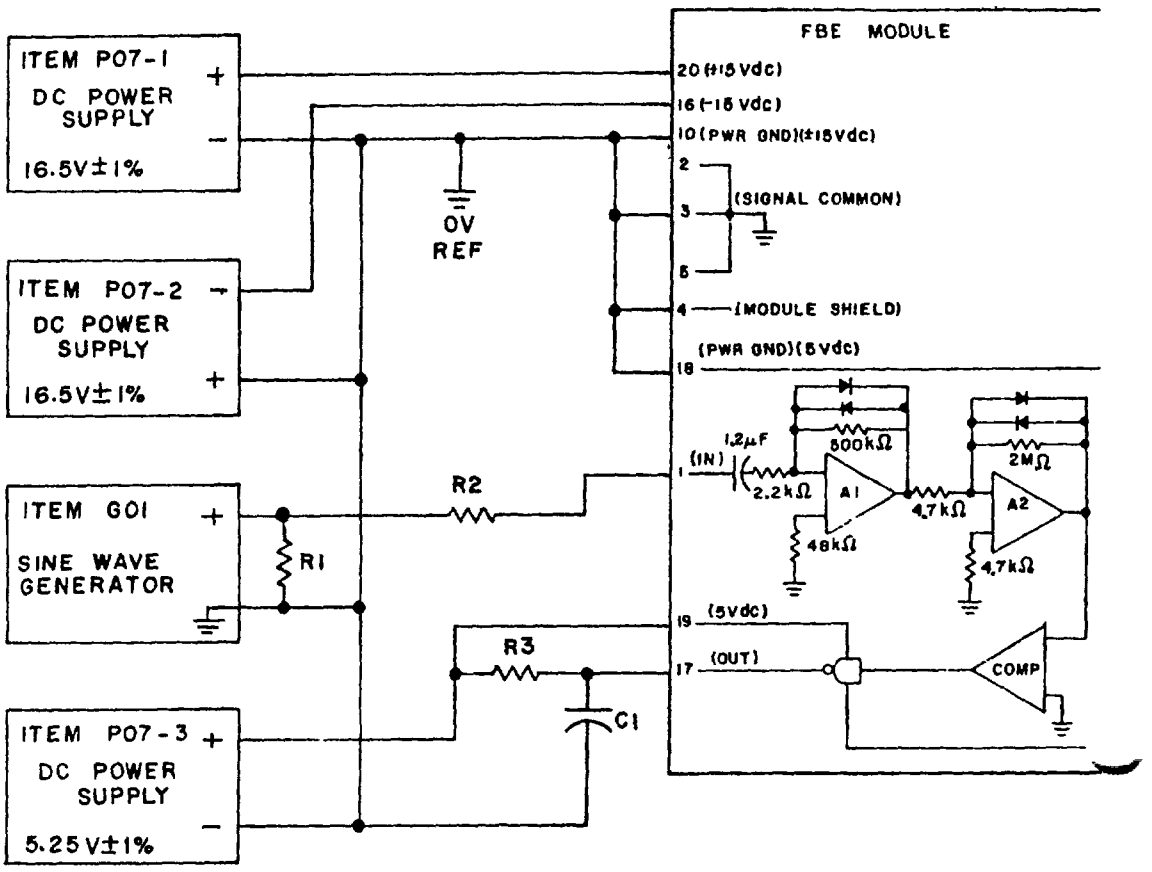
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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
K1-K2	Output symmetry check	X	X	-25	+25	-25	+25	-25	+25	-25	+25	mV
L	Noise	XI	S	-∞	-40	-∞	-40	-∞	-40	-∞	-40	$dBV/\sqrt{Hz}$
M1-M3	Continuity	XII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
N1	Power supply filter capacitor check, 5V	XIII	S	0.42	0.57	0.42	0.57	0.42	0.57	0.42	0.57	V
N2-N3	Power supply filter capacitor check, ±15 V	XIII	S	0.58	0.70	0.58	0.70	0.58	0.70	0.58	0.70	V
P1-P3	Power supply transient amplitude	XIV	Q	0	300	0	300	0	300	0	300	mV
Q	Crosstalk check	XV	Q	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	%
R	Power supply sequence	XVI	Q	← See table XVI →								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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NOTES:
1. R1 = R2 = 600Ω ±5%, 1 W.
2. R3 = 499Ω ±1%, 1/8 W.
3. C1 = 100 pF ±20%.
4. Item G01 setup:
   Amplitude = 1 Vrms ±1%.
   Frequency = 10 kHz ±10%.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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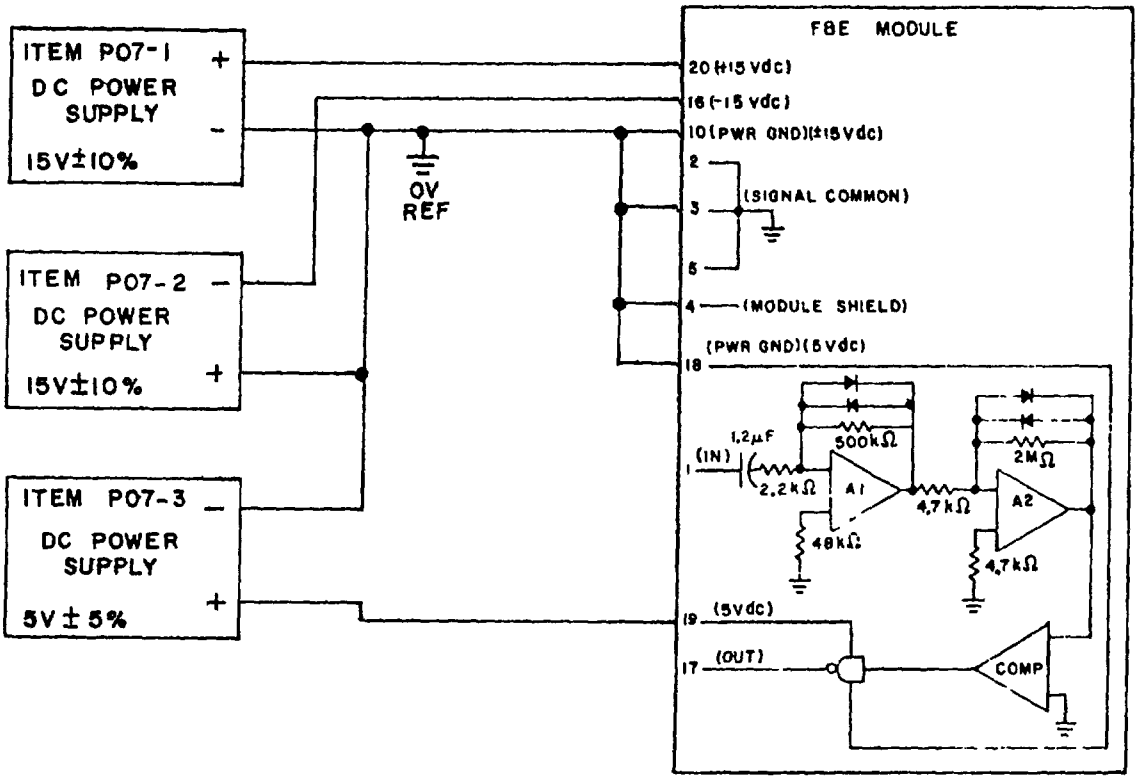


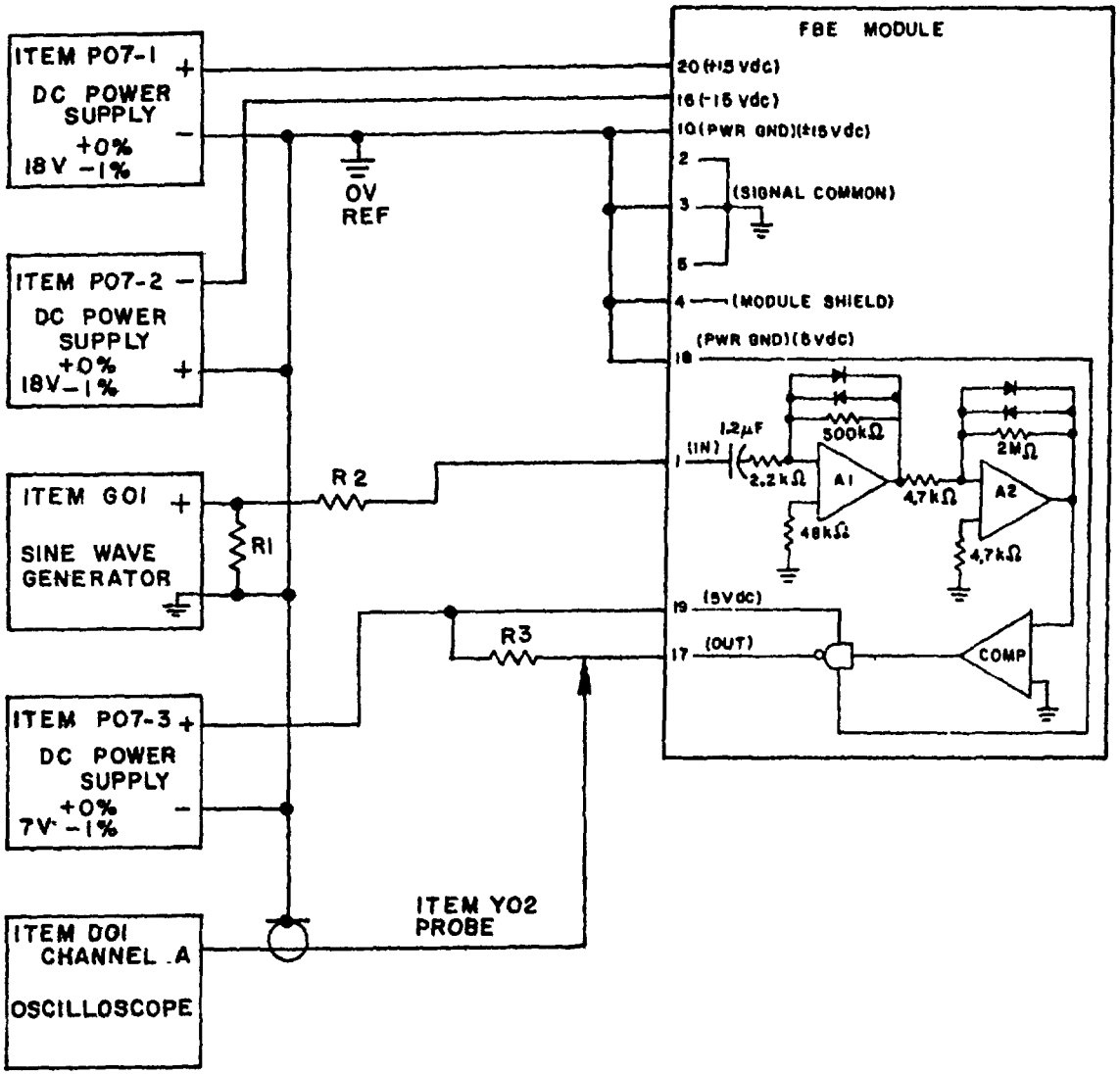
FIGURE 3. Vibration test circuit.

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TABLE II. V_S stress output signal test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
A	V_S stress, output signal	900 (See note 4 of fig 4)	1100	Hz	4

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NOTES:
1. R1 = R2 = 600Ω ±5%, 1 W.
2. R3 = 499Ω ±1%, 1/8 W.
3. Item G01 setup:
   Amplitude = 1 Vrms ±10%.
   Frequency = 1 kHz ±10%.
4. During test, the output shall be a 0 to +7 V (voltage levels for reference only) signal that follows the frequency of the input signal generator.
5. Measurement shall be taken after power has been applied for at least 1 minute.

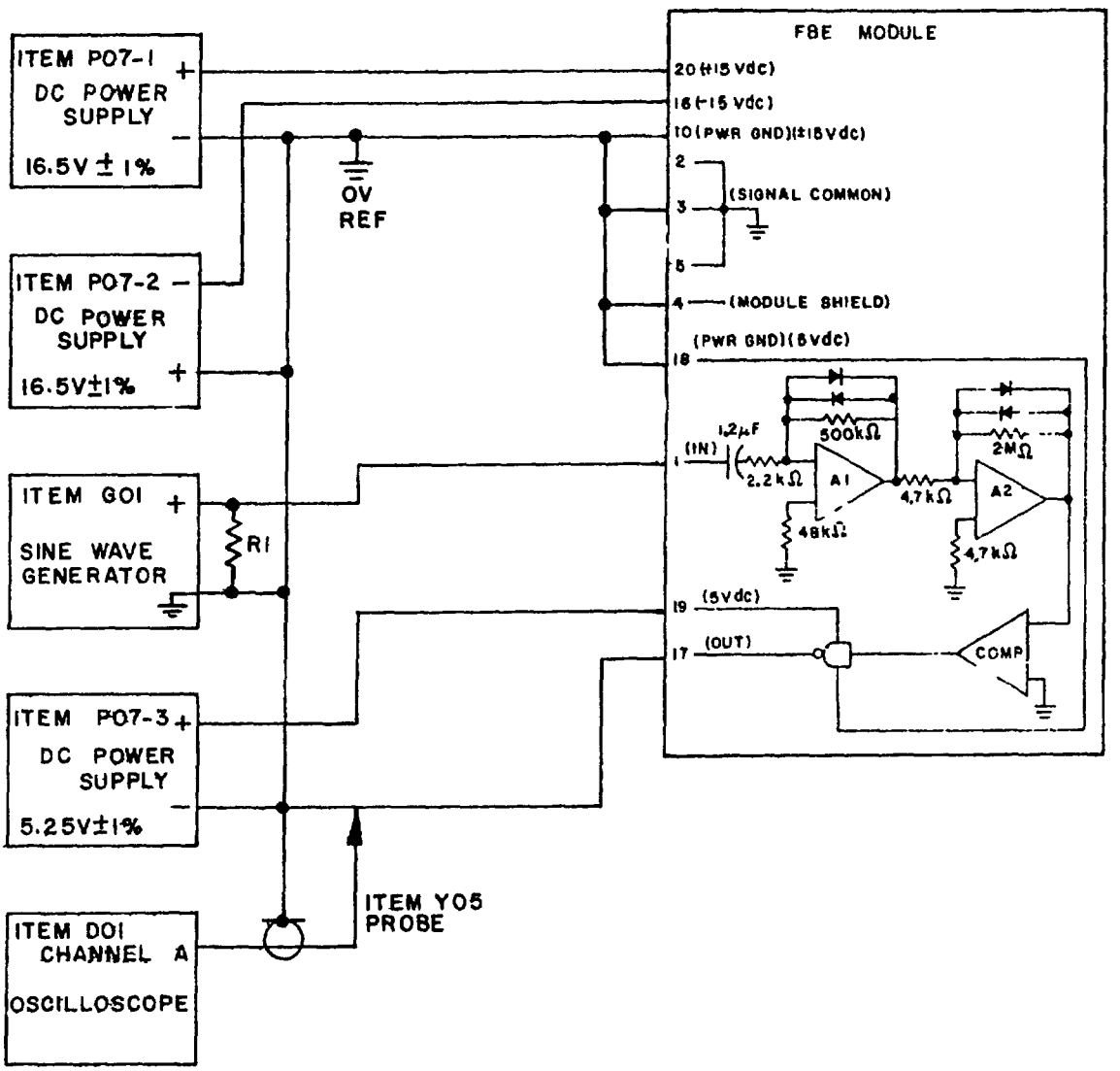
* FIGURE 4. V_S stress output signal test circuit for table II.

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TABLE III. Output short circuit current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
B	Output short circuit current	20	100	mA	5, 6

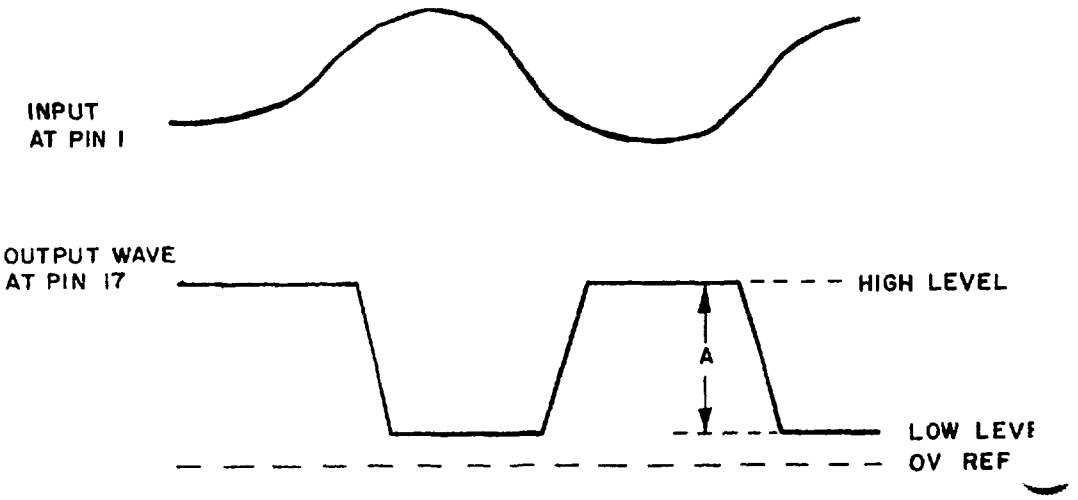
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. Item G01 setup:
   Amplitude = 1 Vrms ±10%.
   Frequency = 1 kHz ±10%.
3. The output shall be short circuited for a maximum of 5 seconds, during which time the measurement shall be made.

FIGURE 5. Output short circuit current test circuit for table III.

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NOTES:
1. All voltage levels are referenced to ground.
2. A = Measure of maximum current.

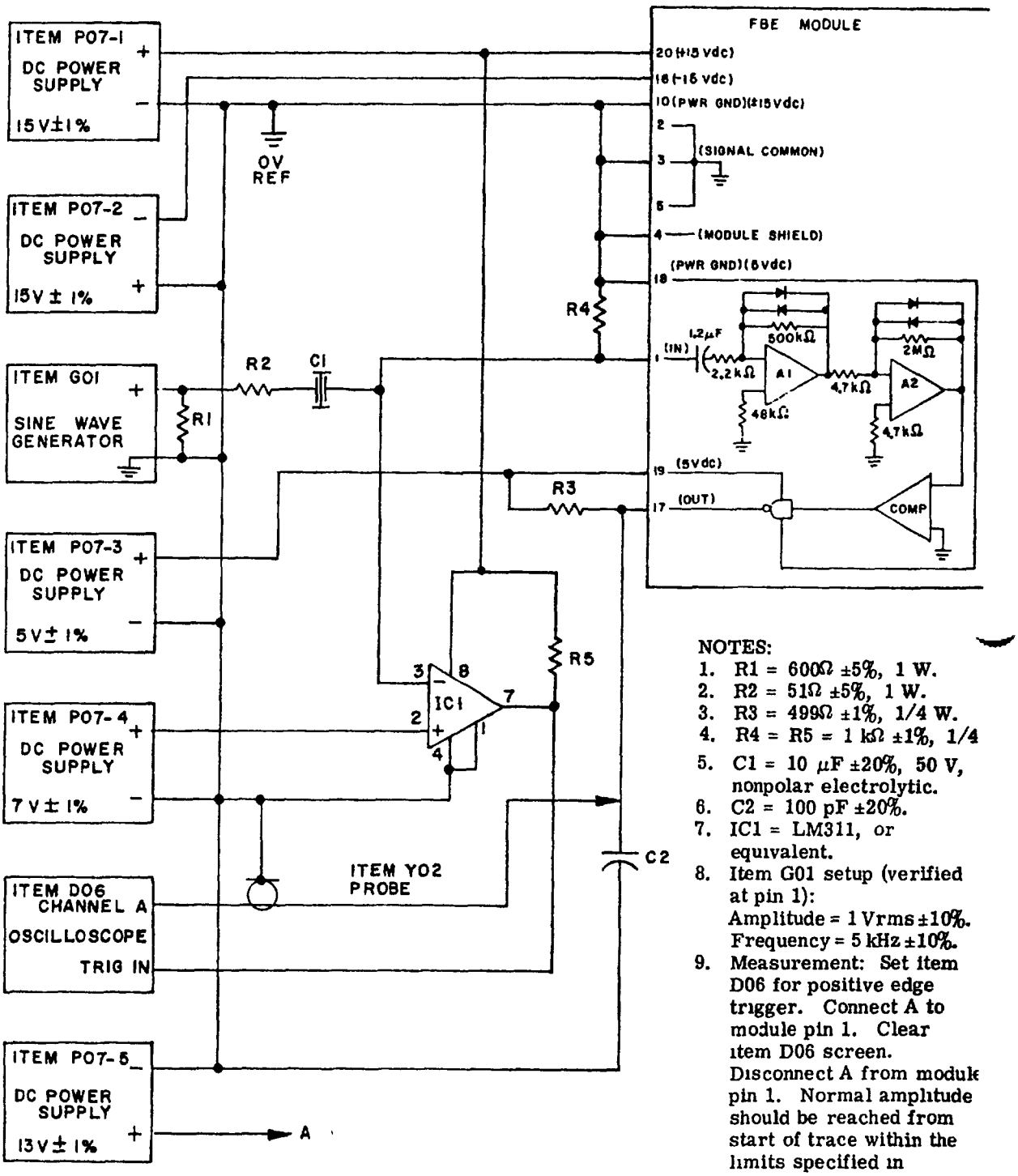
FIGURE 6. Output short circuit current test circuit waveforms for table III.

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TABLE IV. Recovery time test.

Test condition	Parameter	Initial limits 25° +5°C Ambient		Unit	Test figure
		Min	Max		
C	Recovery time	0	100	ms	7

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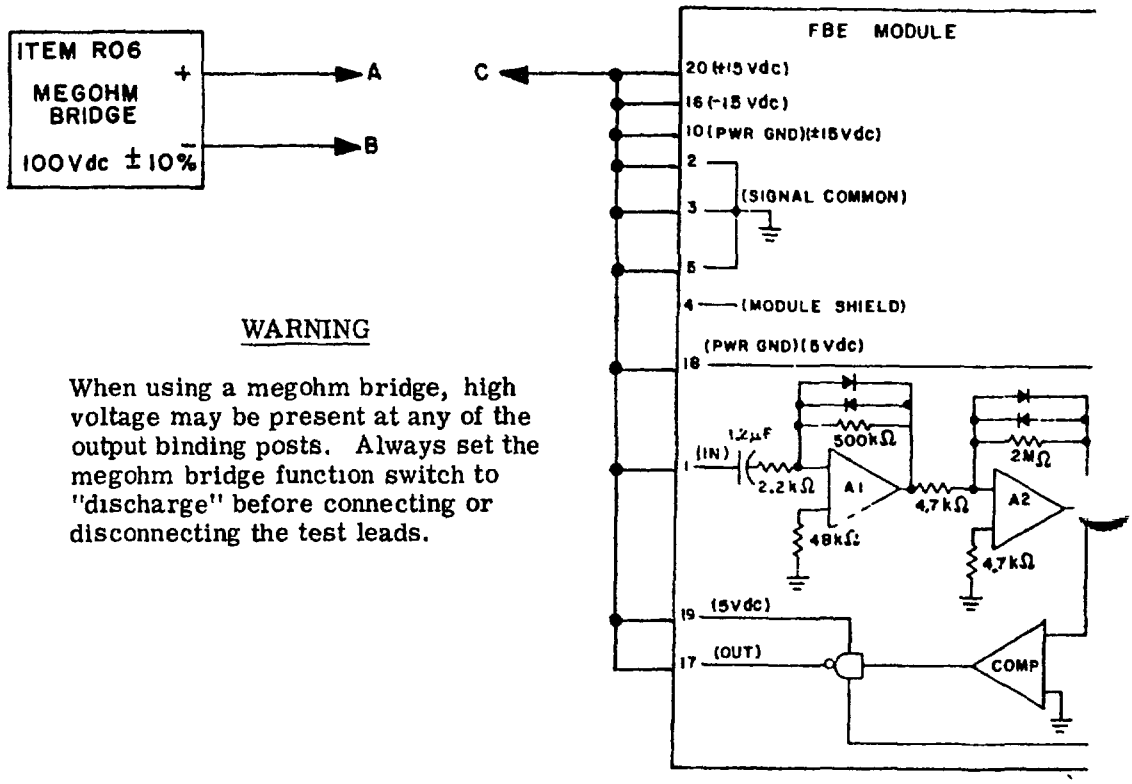
NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 51Ω ±5%, 1 W.
3. R3 = 499Ω ±1%, 1/4 W.
4. R4 = R5 = 1 kΩ ±1%, 1/4
5. C1 = 10 μF ±20%, 50 V, nonpolar electrolytic.
6. C2 = 100 pF ±20%.
7. IC1 = LM311, or equivalent.
8. Item G01 setup (verified at pin 1):
   Amplitude = 1 Vrms ±10%.
   Frequency = 5 kHz ±10%.
9. Measurement: Set item D06 for positive edge trigger. Connect A to module pin 1. Clear item D06 screen. Disconnect A from module pin 1. Normal amplitude should be reached from start of trace within the limits specified in table IV.

FIGURE 7. Recovery time test circuit for table IV.

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Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Isolation, frame pin to module pins	C,6,7,8,9,11, 12,13,14,15	4	10	∞	MΩ	8
D2	Isolation, used to unused pins	C	6,7,8,9,11,12, 13,14,15				
D3	Isolation, unused to unused pins	6	7,8,9,11,12, 13,14,15				
D4		7	8,9,11,12,13, 14,15				
D5		8	9,11,12,13,14, 15				
D6		9	11,12,13,14,15				
D7		11	12,13,14,15				
D8		12	13,14,15				
D9		13	14,15				
D10	Isolation, unused to unused pins	14	15				
D11	Isolation, required to optional pins	1 through 20	21 through 40	10	∞	MΩ	8

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 8. Isolation test circuit for table IV.

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TABLE VI. Output signal test.

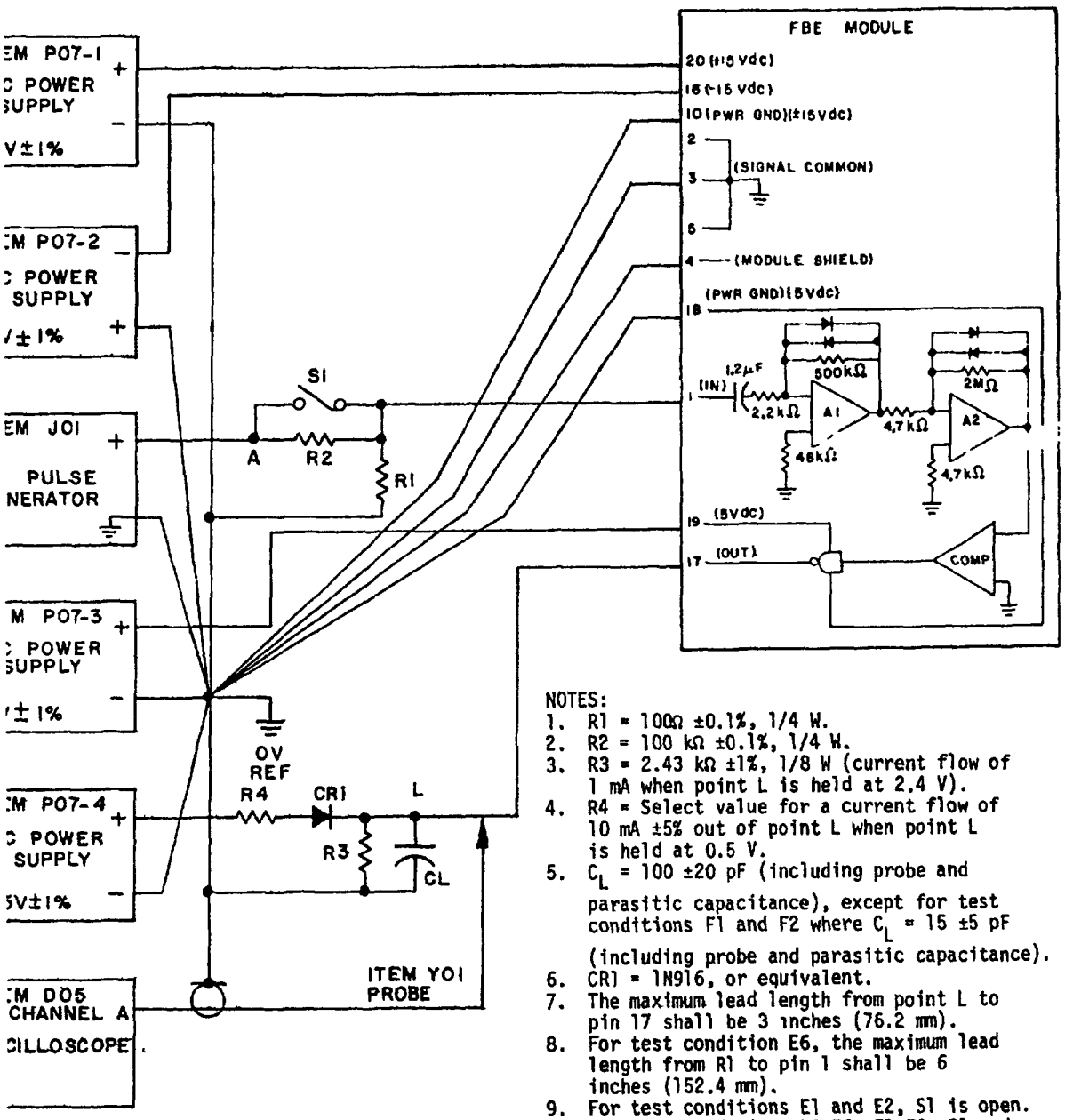
Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
E1	Output signal, voltage, low level (minimum input)	0.0	0.5	V	9, 10
E2	Output signal, voltage, high level (minimum input)	2.5	5.0	↓	↓
E3	Output signal, voltage, low level (maximum input)	0.0	0.5	↓	↓
E4	Output signal, voltage, high level (maximum input)	2.5	5.0	V	↓
E5	Output signal, 10 kHz frequency check	96	104	µs	↓
E6	Output signal, small signal bandwidth check (500 kHz)	1.8	2.2	µs	9, 10

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TABLE VII. Transition time and propagation delay time test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
F1	Transition time (t_{TLH})	18	70	ns	9, 10
F2	Transition time (t_{THL})	5	70		
F3	Transition time (t_{TLH})	30	100		
G1	Propagation delay time (t_{PLH})	100	300		
G2	Propagation delay time (t_{PHL})	100	300	ns	9, 10

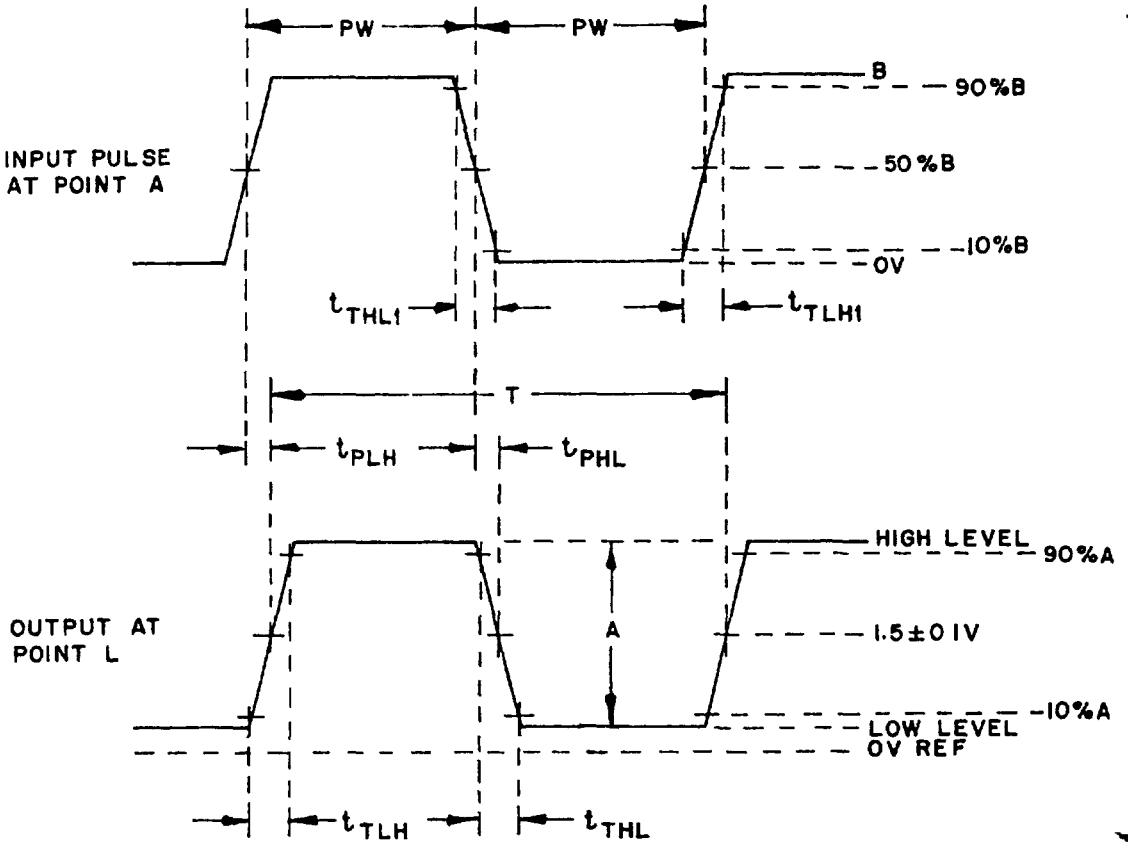
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NOTES:
1. R1 = 100Ω ±0.1%, 1/4 W.
2. R2 = 100 kΩ ±0.1%, 1/4 W.
3. R3 = 2.43 kΩ ±1%, 1/8 W (current flow of 1 mA when point L is held at 2.4 V).
4. R4 = Select value for a current flow of 10 mA ±5% out of point L when point L is held at 0.5 V.
5. C_L = 100 ±20 pF (including probe and parasitic capacitance), except for test conditions F1 and F2 where C_L = 15 ±5 pF (including probe and parasitic capacitance).
6. CR1 = 1N916, or equivalent.
7. The maximum lead length from point L to pin 17 shall be 3 inches (76.2 mm).
8. For test condition E6, the maximum lead length from R1 to pin 1 shall be 6 inches (152.4 mm).
9. For test conditions E1 and E2, S1 is open.
10. For test conditions E3-E6, F1-F3, G1 and G2, S1 is closed.
11. During output switching, item P07-4 shall have a maximum overshoot of 500 mV.
12. Connect the frame of the FBE mating female connector to 0 V reference.
13. All connections to 0 V reference shall terminate at one point as shown.

FIGURE 9. Output signal, transition time, and propagation delay time test circuit for tables VI and VII.

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NOTES:
1. Input pulse parameters for test conditions E1-E5, F, and G:
   (a) PW = 50 ±2 μs.
   (b) $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Input pulse parameters for test condition E6:
   (a) PW = 1.0 ±0.1 μs.
   (b) $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
3. For test conditions E1 and E2, B = 1.5 ±0.1 V.
4. For test conditions E3 and E4, B = 8.5 V ±5%.
5. For test conditions F and G, B = 100 mV ±5%.
6. For test conditions E5 and E6, measure T and compare to the limits specified in table VI.
7. All voltage levels are referenced to ground.
8. For test conditions E5 and E6, B = 1.5 ±0.1 V.

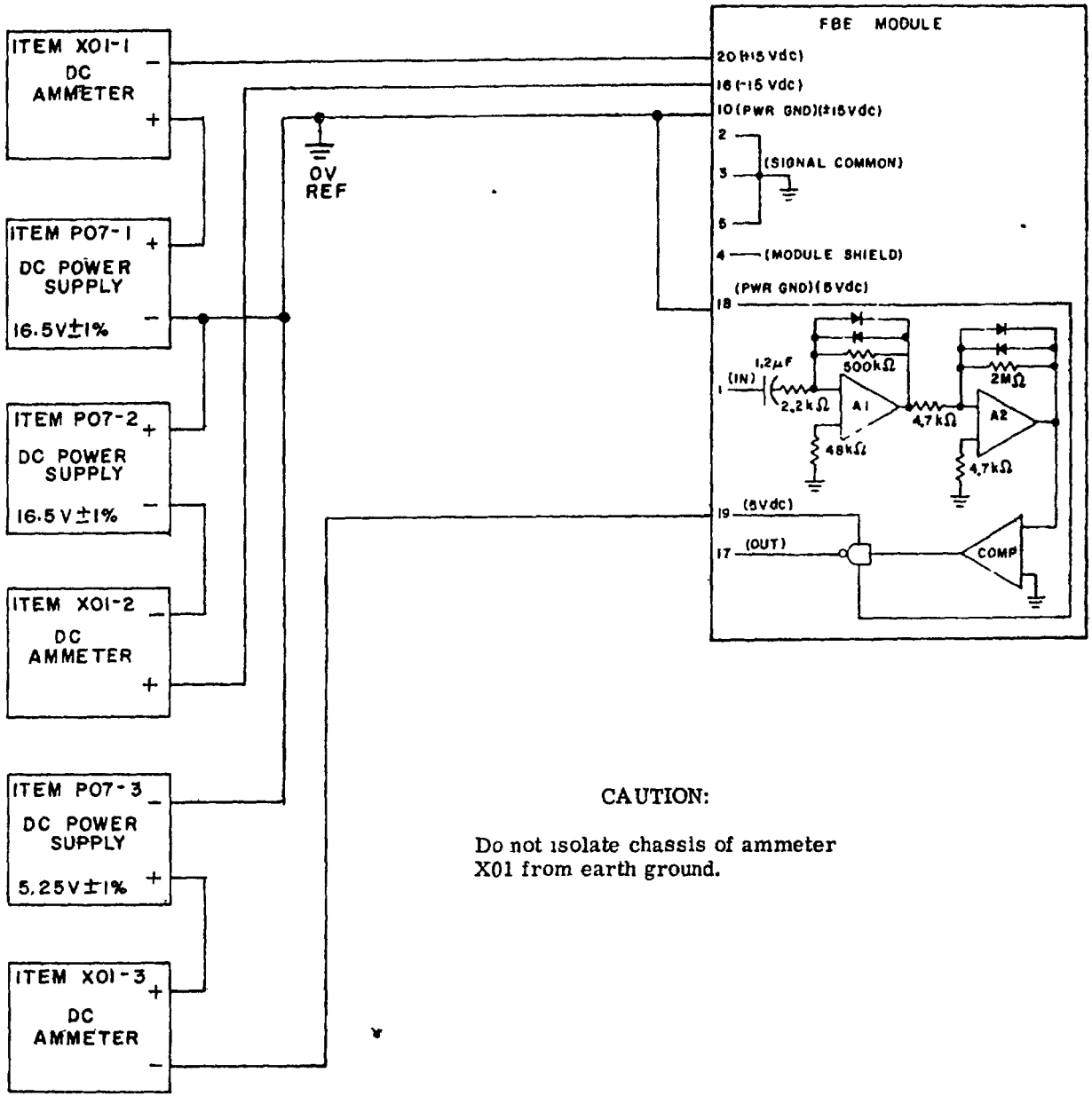
FIGURE 10. Output signal, transition time, and propagation delay time test circuit waveforms for tables VI and VII.

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TABLE VIII. Power supply current tests.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
H1	Power supply current, + 15 V	10	50	mA	11
H2	Power supply current, - 15 V	10	50	mA	11
H3	Power supply current, 5 V	2	20	mA	11

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CAUTION:

Do not isolate chassis of ammeter X01 from earth ground.

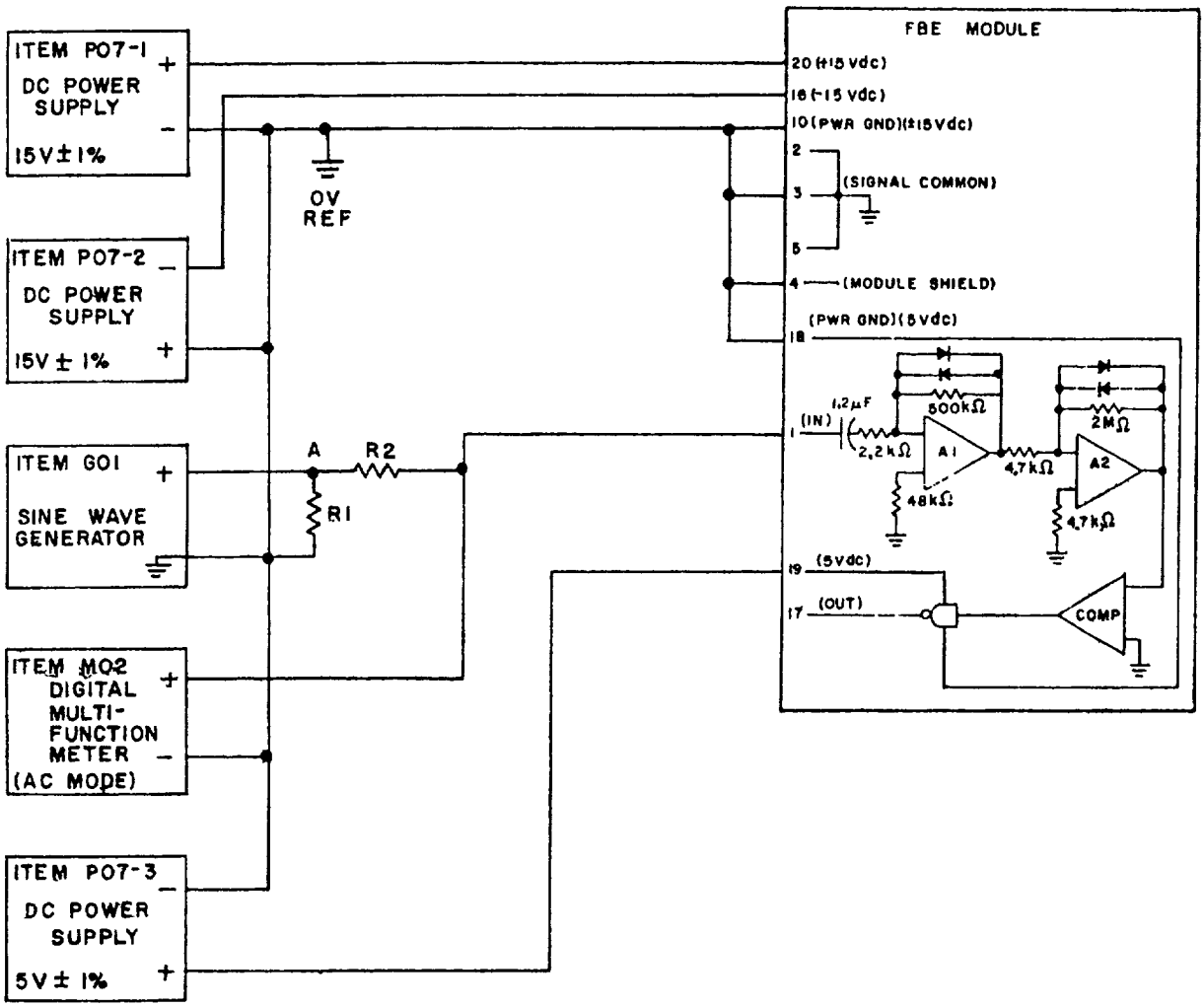
FIGURE 11. Power supply current test circuit for table VIII.

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TABLE IX. Input impedance check test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
J	Input impedance check	0.50	1.01	Vrms	12

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 2 kΩ ±1%, 1/8 W.
3. Item G01 setup verified at point A:
   Amplitude = 1 Vrms ±1%.
   Frequency = 5 kHz ±10%.

FIGURE 12. Input impedance check test circuit for table IX.

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TABLE X. Output symmetry check test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
K1	Output symmetry check	-25	25	mV	13
K2	Output symmetry check	-25	25	mV	13

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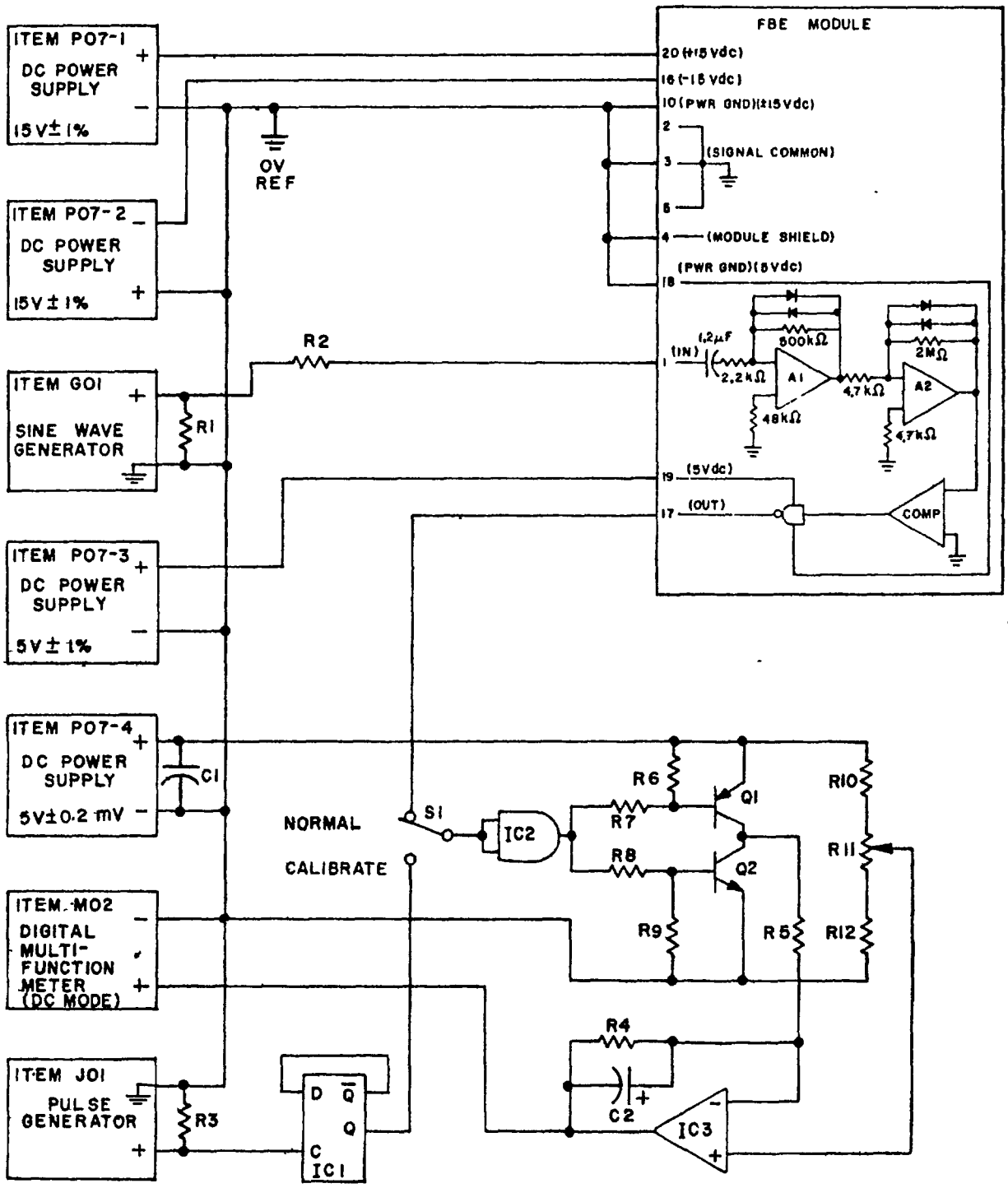


FIGURE 13. Output symmetry check test circuit for table X.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 51Ω ±5%, 1 W.
3. R4 = R5 = 100 kΩ ±0.1%, 1/4 W.
4. R6 = R8 = 1 kΩ ±1%, 1/4 W.
5. R7 = 4.7 kΩ ±1%, 1/4 W.
6. R9 = 10 kΩ ±1%, 1/8 W.
7. R10 = 105 kΩ ±1%, 1/8 W.
8. R11 = 10 kΩ, 1/8 W potentiometer.
9. R12=26.1kΩ ±1%, 1/8 W.
10. C1 = 0.1 μF ±10%.
11. C2 = 5 μF ±1%, 20 V, electrolytic.
12. Q1 = 2N2907A, or equivalent.
13. Q2 = 2N2222A, or equivalent.
14. IC1 = SN7474, or equivalent.
15. IC2 = SN7408, or equivalent.
16. IC3 = μA741, or equivalent.
17. Test procedure for test condition K1:
    (a) Item G01 setup:
        Amplitude = 100 mVrms ±1%.
        Frequency = 100 ±1 Hz.
    (b) Item J01 setup:
        Amplitude = 0 to 5 V ±5%.
        Frequency = 200 Hz.
        $t_{TLH} = t_{THL} = 5 \pm 2$ ns.
        Waveform = Square.
    (c) Switch S1 to "calibrate" position.
    (d) Adjust R11 for 0 ±0.5 mV output on item M02.
    (e) Switch S1 to "normal" position for test.
18. Test procedure for test condition K2, repeat as specified in note 17 except as follows:
    (a) Item G01 frequency = 40 kHz ±1%.
    (b) Item J01 frequency = 80 kHz.
19. Capacitance and lead length on pin 17 should be kept as small as possible.

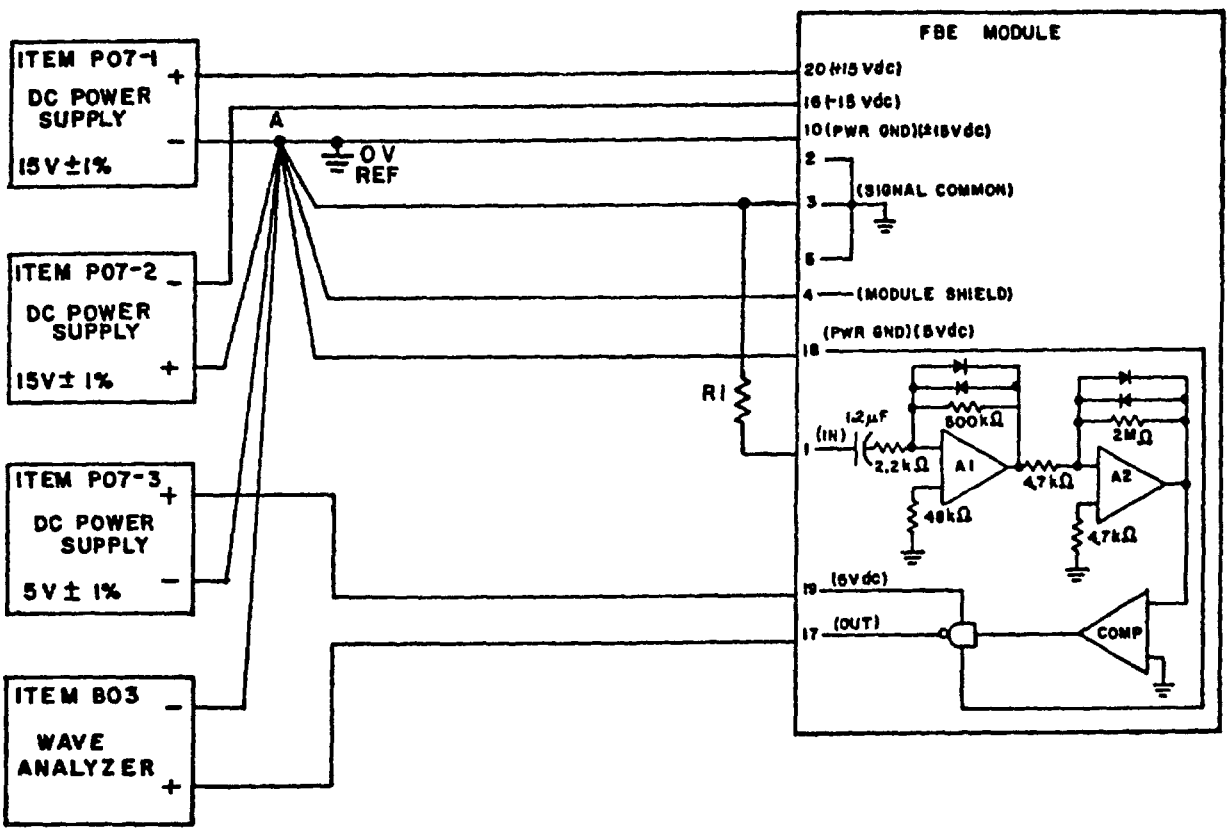
FIGURE 13 Output symmetry check test circuit for table X - Continued.

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TABLE XI. Noise test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
L	Noise (20 Hz to 50 kHz)	-∞	-40	dBV/$\sqrt{Hz}$	14

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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. Item B03 shall be swept through the frequency range of 20 Hz to 50 kHz and at no point in this frequency range shall item B03 read more than 0.010 Vrms (-40 dBV).
3. Adjust item B03 to 1 Hz bandwidth or perform conversion calculation from wider bandwidth as specified by wave analyzer manufacturer.
4. 0 dB = 1.0 V.
5. All connections to point A shall terminate at one point as shown. This point shall be at the common of the item P07-1 power supply.

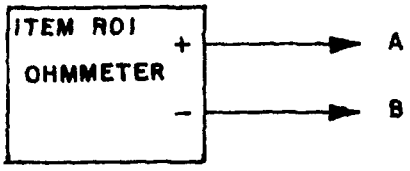
FIGURE 14. Noise test circuit for table XI.

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TABLE XII. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Continuity	4	Module cover	0.0	1.0	Ω	15
M2	Continuity	2	3	0.0	1.0	Ω	15
M3	Continuity	2	5	0.0	1.0	Ω	15

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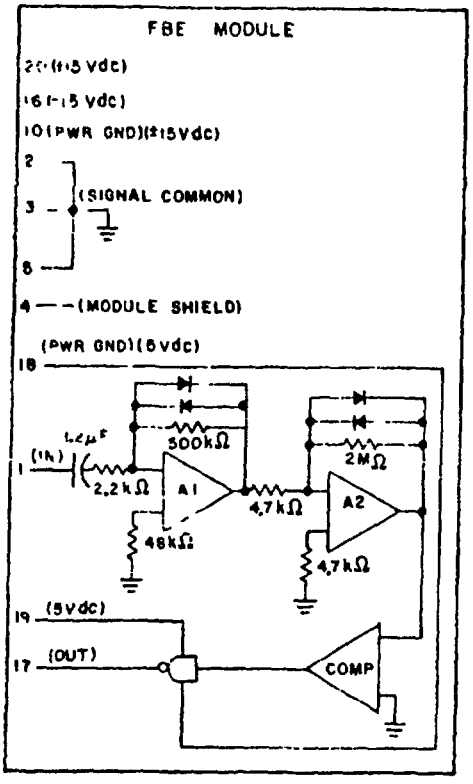


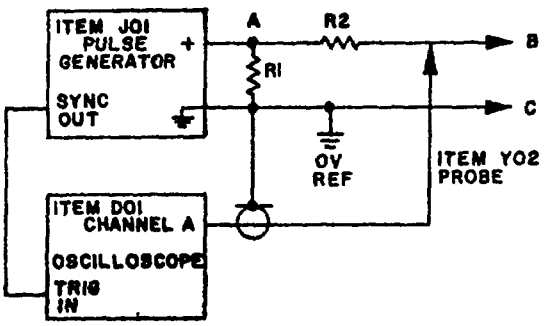
FIGURE 15. Continuity test circuit for table XII.

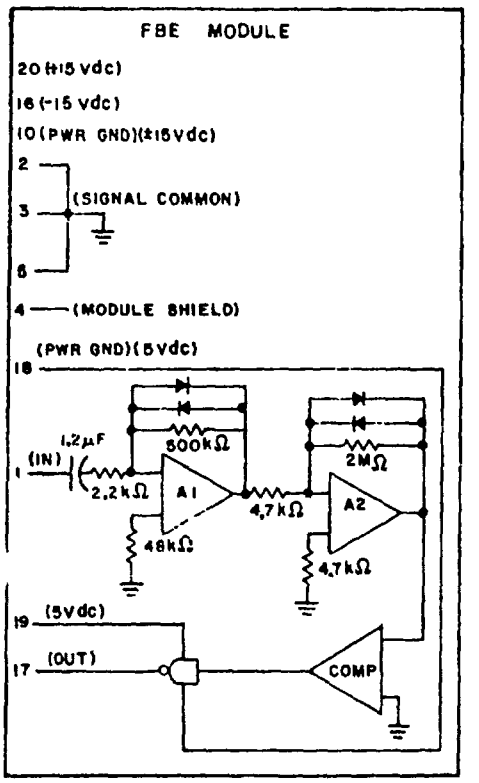
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TABLE XIII. Power supply filter capacitor check test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		B to pin	C to pin	Min	Max		
N1	Power supply filter capacitor check, 5 V	19	18	0.42	0.57	V	16, 17
N2	Power supply filter capacitor check + 15 V	20	10	0.58	0.70	V	16, 17
N3	Power supply filter capacitor check - 15 V	10	16	0.58	0.70	V	16, 17

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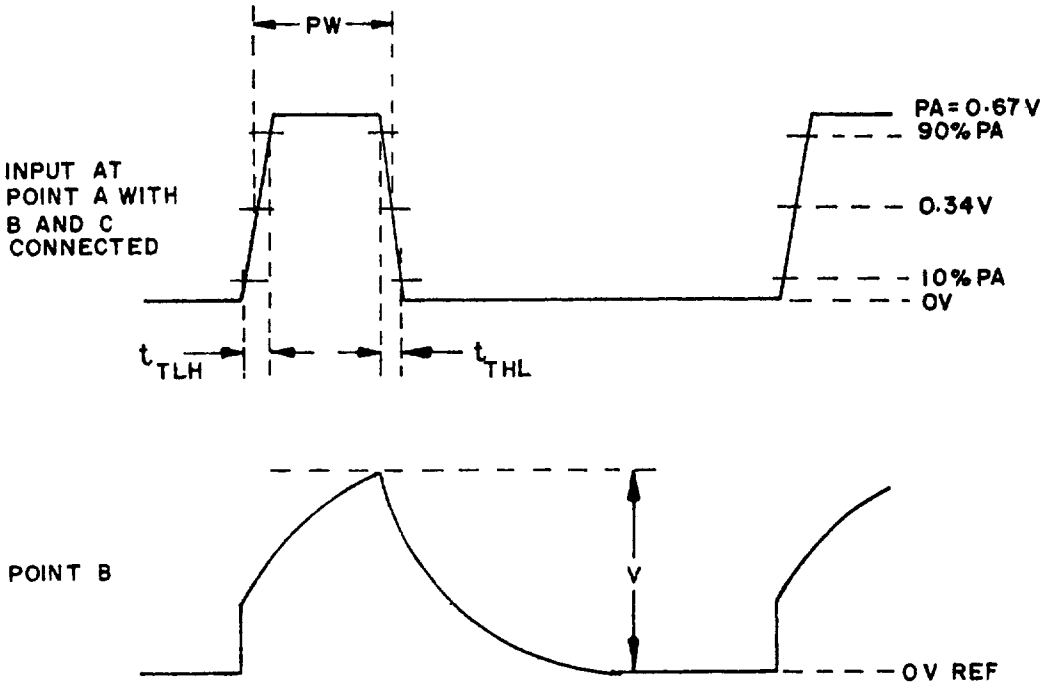




NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 100Ω ±1%, 1/8 W.

FIGURE 16. Power supply filter capacitor check test circuit for table XIII.

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NOTES:
1. Input pulse parameters for test condition N1:
   (a) PW = 14.0 ±0.5 μs.
   (b) PRP = 200 ±20 μs.
   (c) $t_{TLH} = t_{THL}$ = 25 ±10 ns.
2. Input pulse parameters for test conditions N2 and N3:
   (a) PW = 570 ±20 μs.
   (b) PRP = 7.0 ±0.5 ms.
   (c) $t_{TLH} = t_{THL}$ = 25 ±10 ns.
3. V = measurement from reference to peak.

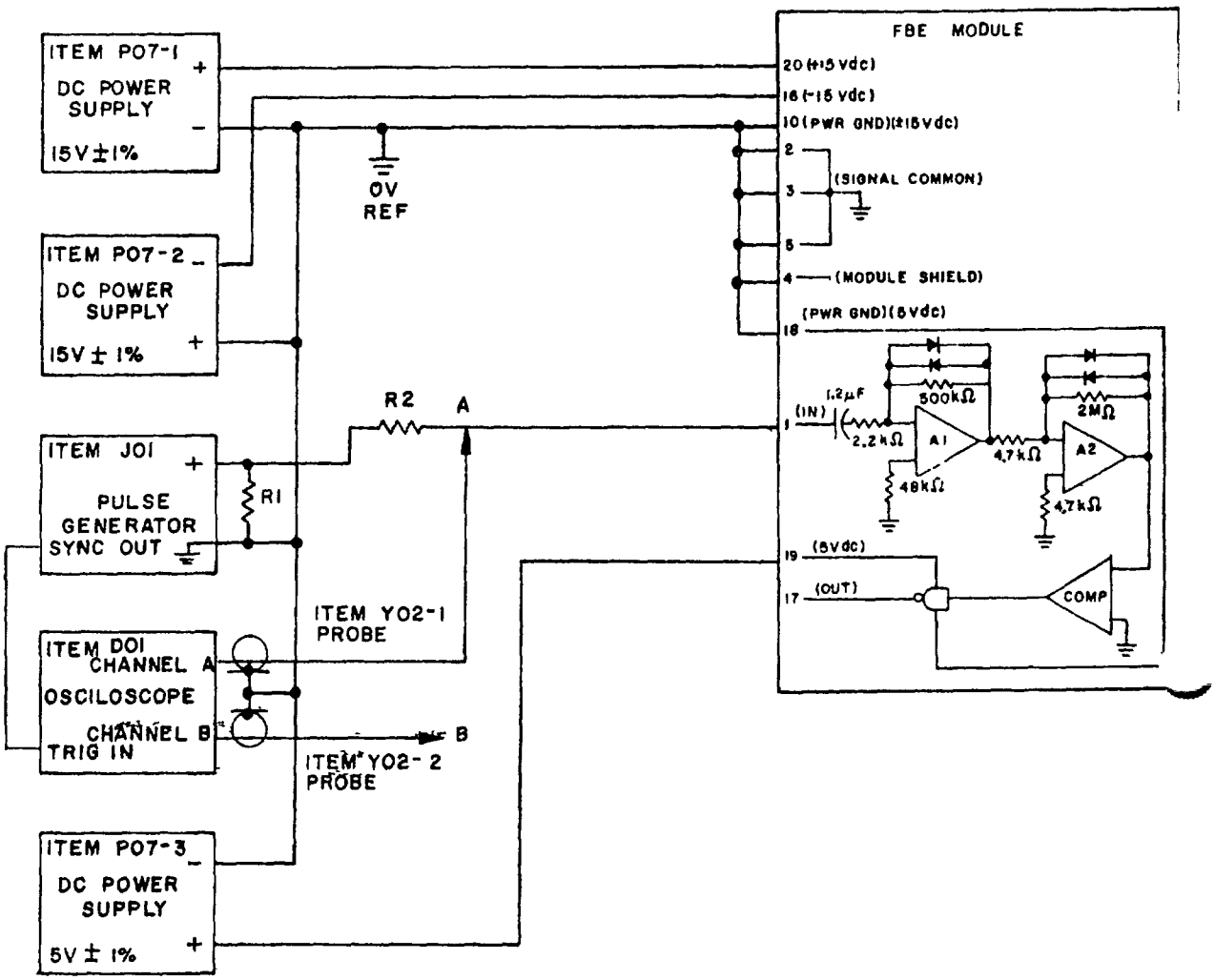
FIGURE 17. Power supply filter capacitor check test circuit waveforms for table XIII.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		B to pin	Min	Max		
P1	Power supply transient amplitude, + 15 V	20	0	300	mV	18, 19
P2	Power supply transient amplitude, - 15 V	16	0	300	mV	18, 19
P3	Power supply transient amplitude, 5 V	19	0	300	mV	18, 19

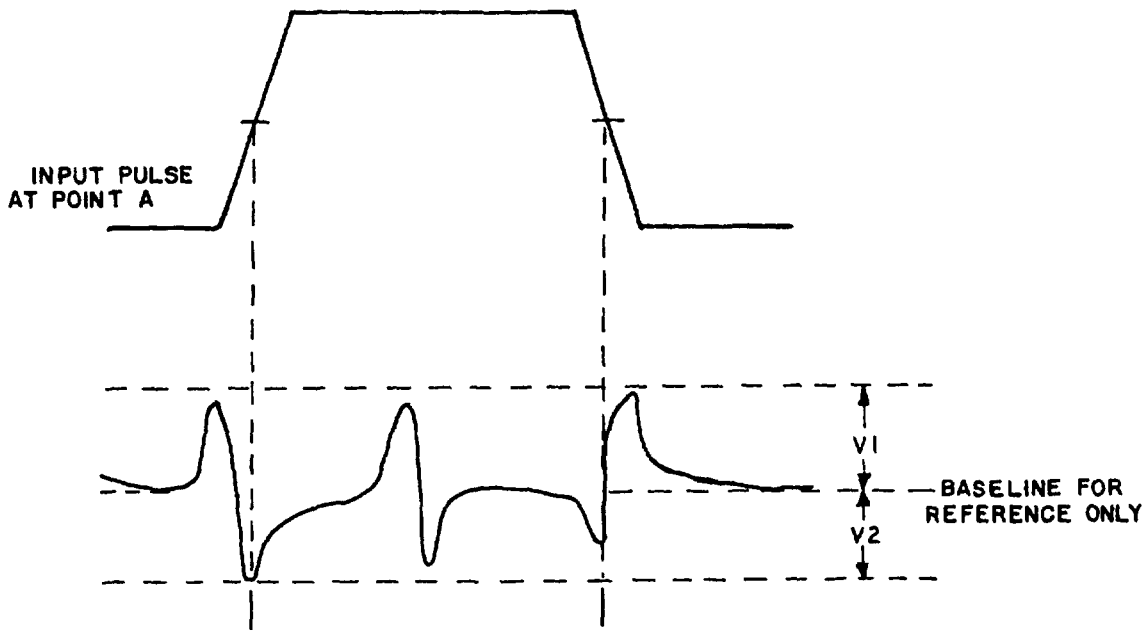
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Item J01 setup:
   Amplitude = 0 to 1 V ±1%.
   Frequency = 1 kHz ±10%.
   Waveform = Square.

FIGURE 18. Power supply transient amplitude test circuit for table XIV.

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NOTES:
1. For test conditions P1 and P3, measure V2 = maximum amplitude of most negative peak.
2. For test condition P2, measure V1 = maximum amplitude of most positive peak.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIV.

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TABLE XV. Crosstalk check test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Q	Crosstalk check	0.0	1.0	%	20

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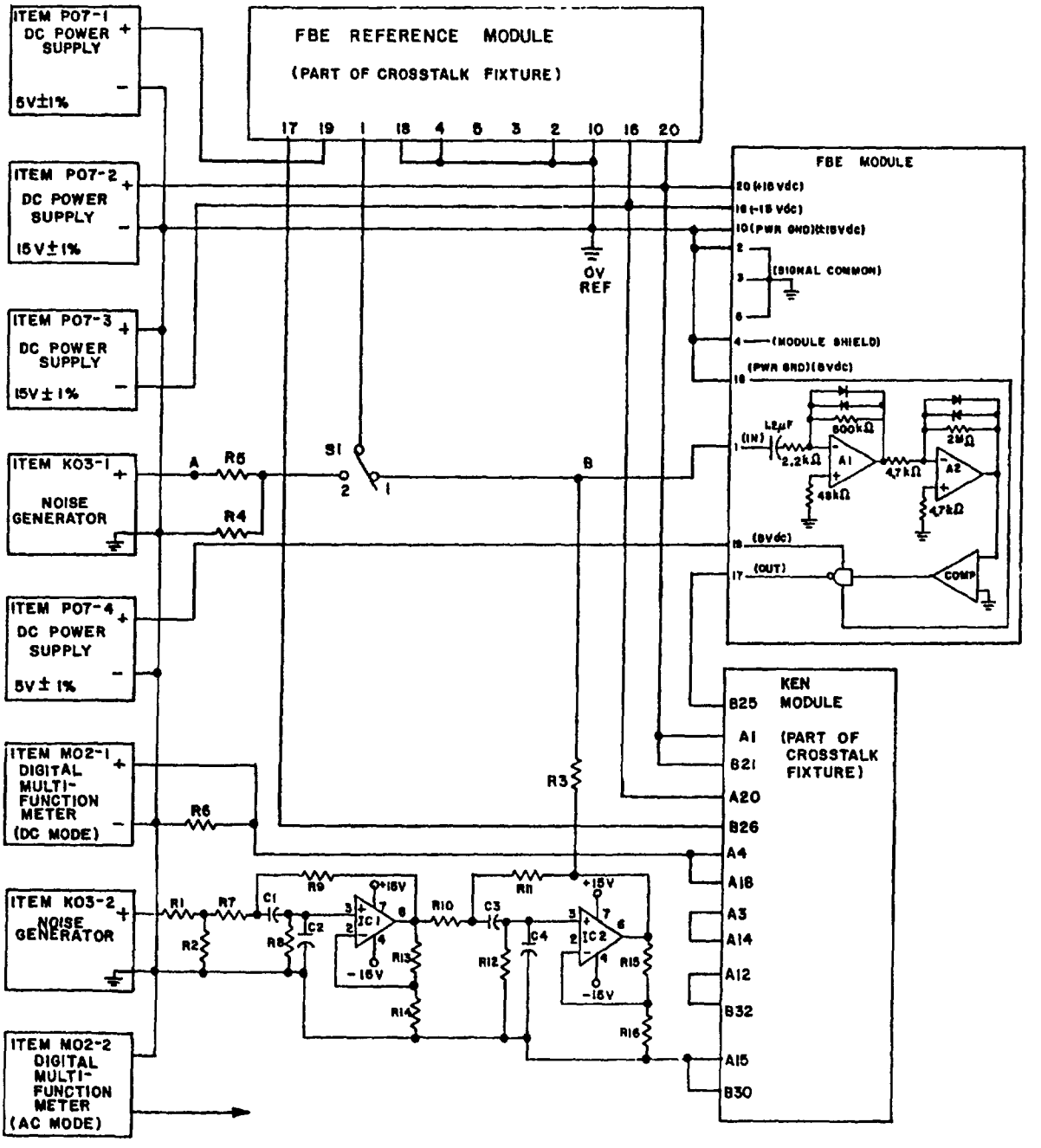


FIGURE 20. Crosstalk check test circuit for table XV.

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NOTES:
1. R1 = R2 = 1 kΩ ±1%, 1/8 W.
2. R3 = R4 = 49.9Ω ±1%, 1/8 W.
3. R5 = R6 = 49.9 kΩ ±1%, 1/8 W.
4. R7-R12 = 15 kΩ ±1%, 1/8 W.
5. R13-R16 = 50 kΩ ±1%, 1/8 W.
6. C1-C4 = 2,000 pF ±20%, 50 V.
7. IC1 = IC2 = HA2600, or equivalent.
8. Set frequency mode of item K03 to 5 Hz - 20 kHz.
9. Use item M02-2 to set the level at points A and B = 1 Vrms ±1%.
10. The FBE reference module shall have passed all other tests of this specification and shall have been tested for crosstalk acceptability against the correlation module(s).
11. During this test, the FBE reference module and the FBE test module shall be positioned side-by-side such that they are touching each other.
12. The FBE reference module and the KEN module shall be permanently affixed to the crosstalk test fixture.
13. Place S1 in position 1 and record item M02-1 reading. Place S1 in position 2. The item M02-1 reading in position 2 shall be less than 1% of the position 1 reading.

FIGURE 20. Crosstalk check test circuit for table XV - Continued.

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TABLE XVI. Power supply sequence test.

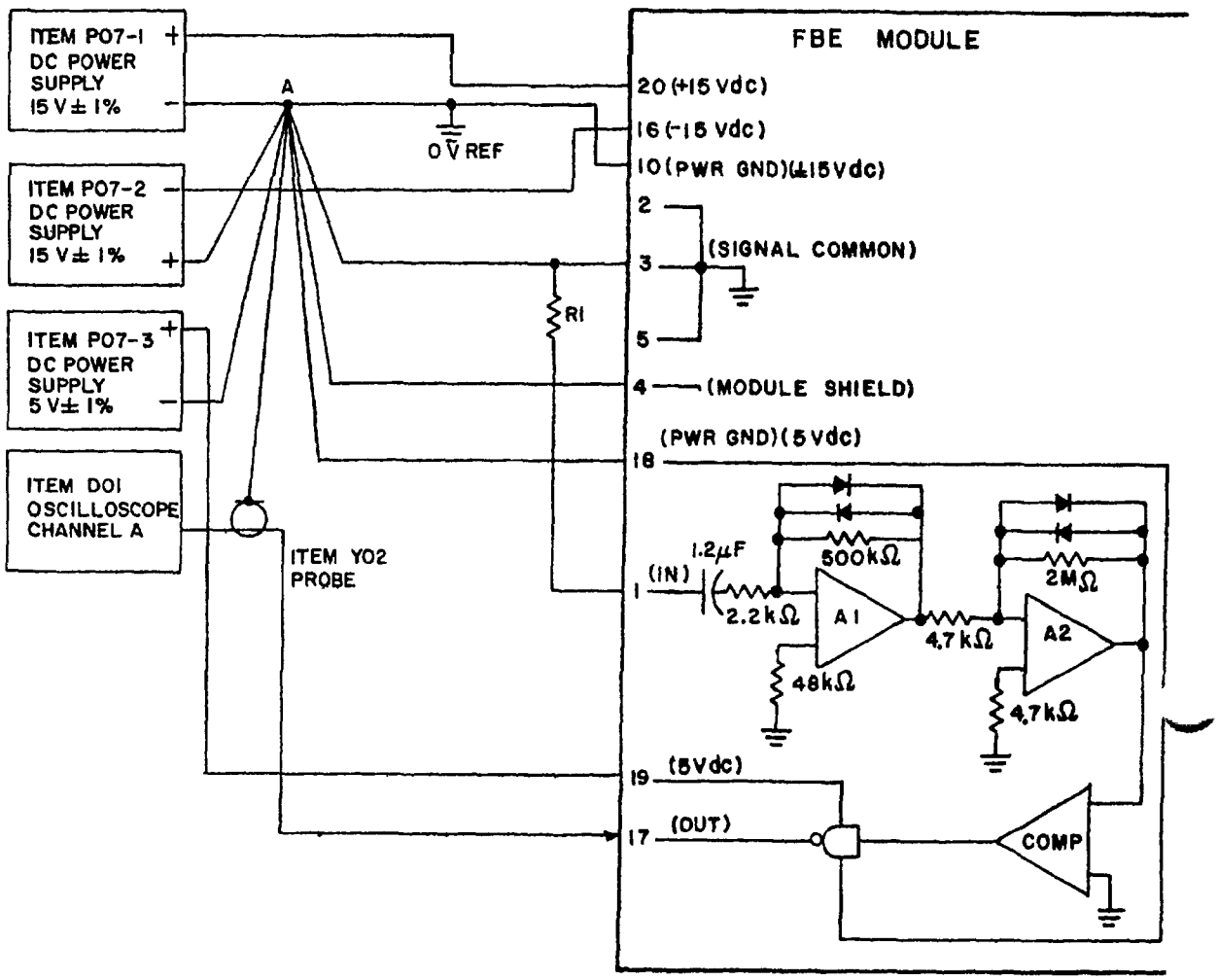
Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
R1	Power supply sequence	See note 1			21
R2	Power supply sequence	See note 2			21

NOTES:

1. For test condition R1, the waveform shall either be noise which does not have a consistent amplitude and period or shall approximate a sine wave (note the waveform which is observed).

2. For test condition R2, the waveform shall not change from the waveform of test condition R1. Specifically the waveform shall not be a fast rise time, distinct square wave with amplitude as specified in test conditions E1 and E2.

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NOTES:
1. R1 = 100Ω ±1%, 1/8 W.
2. All connections to part A shall terminate at one point (-terminal of item P07-1) as shown.
3. For test condition R1. Energize the module by turning the power supplies on in the following sequence; item P07-3, item P07-2 and item P07-1. After all power supplies are on, the waveform displayed on item D01 shall meet the requirements of test condition R1.
4. For test condition R2: Without de-energizing the module from test condition R1, change the item P07-2 voltage continuously from 15 V to 6.0 $^{+0}_{-0.2}$ V and back to 15 V. The waveform displayed on item D01 after the item P07-2 voltage has been returned to 15 V shall meet the requirements of test condition R2.

FIGURE 21. Power supply sequence test circuit for table XVI.

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